University of Chittagong
Department of Computer Science and Engineering
8th Semester B.Sc. Engg. Examination - 2020
Course Code: CSE 815
Course Title: Machine Learning
Total Marks: 52.5 Marks Time: 4:00 Hours

[Answer the six of the following questions Three from each Section A and Section B; Figures in the right-hand margin indicate full marks.]

Section A

1. (a) Define: 1.75
   (i) input/hidden/output layers
   (ii) weights
   (iii) activation functions
   (iv) error

   (b) Write short notes on: (i) Artificial Neurons, (ii) Artificial Neural Network. 2

   (c) Consider the following training set 3

No.	input values	desired output value
1.	$x^1 = (1, 0, 1)^T$	-1
2.	$x^2 = (0, -1, -1)^T$	1
3.	$x^3 = (-1, -0.5, -1)^T$	1

   Assume the learning rate is 0.1 and the initial weight vector is $w_0 = (1, -1, 0)^T$
   After training draw the neural network.

   (d) Make a comparison among supervised, unsupervised and hybrid ANNs in terms of learning rule, Architecture, learning algorithm and task. 2

2. (a) Draw a flow diagram to demonstrate the Genetic Algorithm process. 2

   (b) Suppose a genetic algorithm uses chromosomes of the form x = abcdefgh with a fixed length of eight gens. Each gene can be any digit between 0 and 9. Let the initial population consist of four individuals with the following chromosomes: 2

   x1 = 6 5 4 1 3 5 3 2
   x2 = 8 7 1 2 6 6 0 1
   x3 = 2 3 9 2 1 2 8 5
   x4 = 4 1 8 5 2 0 9 4

   Perform the following crossover operations:
   (i) Cross the fittest individuals (x1,x2) using one-point crossover at the middle point.
   (ii) Cross the third and fourth fittest individuals using a two-point crossover (points b and f)

   (c) Let function $f(x) = 2x - x^2/16$ be defined on the interval [0, 31]. Show the use of a genetic algorithm for determining the maximum of the function in the given interval. Note that the relative fitness function g(x) is defined as below: 4.75

$$g(x) = \frac{f(x)}{\sum_{x \in p^{(k)}} f(x)}$$

3. (a) Write short notes on Decision tree. 1
   (b) What is CART? Describe four important string problems in pattern recognition. 2
   (c) Write down Boyer-Moore String Matching algorithm and explain with an example. 2.25
   (d) What is Edit Distance? Write down Edit Distance algorithm and explain with an example. 2
   (e) Write short notes on: 1.5
      (i) String matching with Error
      (ii) String matching with the "Don't-Care" Symbol.

4. (a) What is support vector? Why maximum margin in SVM technique? 2
   (b) What is the "kernel trick"? How do you compute M in terms of $\mathbf{W}$ and b? 2.25
   (c) Write algorithm for K-means clustering algorithm. 1.5
   (d) Suppose that the machine learning task is to cluster the following students CGPA into three clusters, i.e., k = 3: 3

A	B	C	D	E	F
3.45	3.78	2.98	3.24	4.0	3.9

The distance function is Euclidean distance. Suppose initially we assign A, C, E as the center of each cluster. Use k-means algorithm to show only the final three clusters.

Section B

5. (a) Briefly describe the application of neural network. 1.25
   (b) Given a two-input neuron with the following parameters: b= 1.2, $\mathbf{W} = [3\ \ 2]$ and $\mathbf{p} = [-5\ \ 6]^T$, calculate the neuron output for the following functions: 1
      i. A symmetrical hard limit transfer function
      ii. A saturating linear transfer function
   (c) A single-layer neural network is to have six inputs and two outputs. The outputs are to be limited to and continuous over the range 0 to 1. What can you tell about the network architecture? Specifically: 1
      i. How many neurons are required?
      ii. What are the dimensions of the weight matrix?
   (d) Describe multiple layers of neurons with necessary figures. 2
   (e) Solve the following classification problem with the perceptron rule. Apply each input vector in order, for as many repetitions as it takes to ensure that the problem is solved. Draw a graph of the problem only after you have found a solution. 3.5

$$\left\{p_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, t_1 = 0\right\}\left\{p_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, t_2 = 1\right\}\left\{p_3 = \begin{bmatrix} -2 \\ 2 \end{bmatrix}, t_3 = 0\right\}\left\{p_4 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, t_4 = 1\right\}$$

Use the initial weights and bias:

$W(0) = [0 \quad 0]$ $\qquad$ $b(0) = 0.$

6. (a) What do you understand by feature space? What is linearly separable and not separable? Explain with example. 2

(b) A produce dealer has a warehouse that stores a variety of fruits and vegetables. When fruit is brought to the warehouse, various types of fruit may be mixed together. The dealer wants a machine that will sort the fruit according to type. There is a conveyer belt on which the fruit is loaded. This conveyer passes through a set of sensors, which measure three properties of the fruit: *shape*, *texture* and *weight*. These sensors are somewhat primitive. The shape sensor will output a 1 if the fruit is approximately round and a -1 if it is more elliptical. The texture sensor will output a 1 if the surface of the fruit is smooth and a -1 if it is rough. The weight sensor will output a 1 if the fruit is more than one pound and a -1 if it is less than one pound. The three sensor outputs will then be input to a neural network. The purpose of the network is to decide which kind of fruit is on the conveyor, so that the fruit can be directed to the correct storage bin. To make the problem even simpler, let's assume that there are only two kinds of fruit on the conveyor: Banana and Pineapple.
As each fruit passes through the sensors it can be represented by a three dimensional vector. The first element of the vector will represent shape, the second element will represent texture and the third element will represent weight as below.

$$p = \begin{bmatrix} shape \\ texture \\ weight \end{bmatrix}$$

The neural network will receive one three-dimensional input vector for each fruit on the conveyer and must make a decision as to whether the fruit is a *banana* (p_1) or a pine*apple* (p_2).

(i) Write down the prototype templates for *banana* (p_1) or a pine*apple* (p_2). 1

(ii) Design a perceptron to recognize these patterns. 1

(iii) Test the operation of your networks by applying several different input patterns. Discuss the advantages and disadvantages of each network. 2.75

(c) Write short notes on Hamming network and Hopfield network. 2

7. (a) What do you understand about Instance-Based Learning? 1

(b) Describe the k-nearest neighbor learning. 2.75

(c) What is a distance metric? What propertles does it have to satisfy? 2
Define (any three)
(i) Minkowski-norm, (ii) Euclidean norm
(iii) Manahattan norm (iv) Hamming distance
(v) Maximum norm

(d) Write down Sequential Covering Algorithm. 2

(e) What is the difference between inductive and analytical learning methods? 1

8. (a) What is Machine learning? Describe the process of designing a learning system. 3

(b) Bayesian classifiers are statistical classifiers. Justify your answer. 1.75

(c) Why Naive Bayesian Classification is known as class conditional independence? 2

(d) What is Q learning? Describe Q learning algorithm. 2